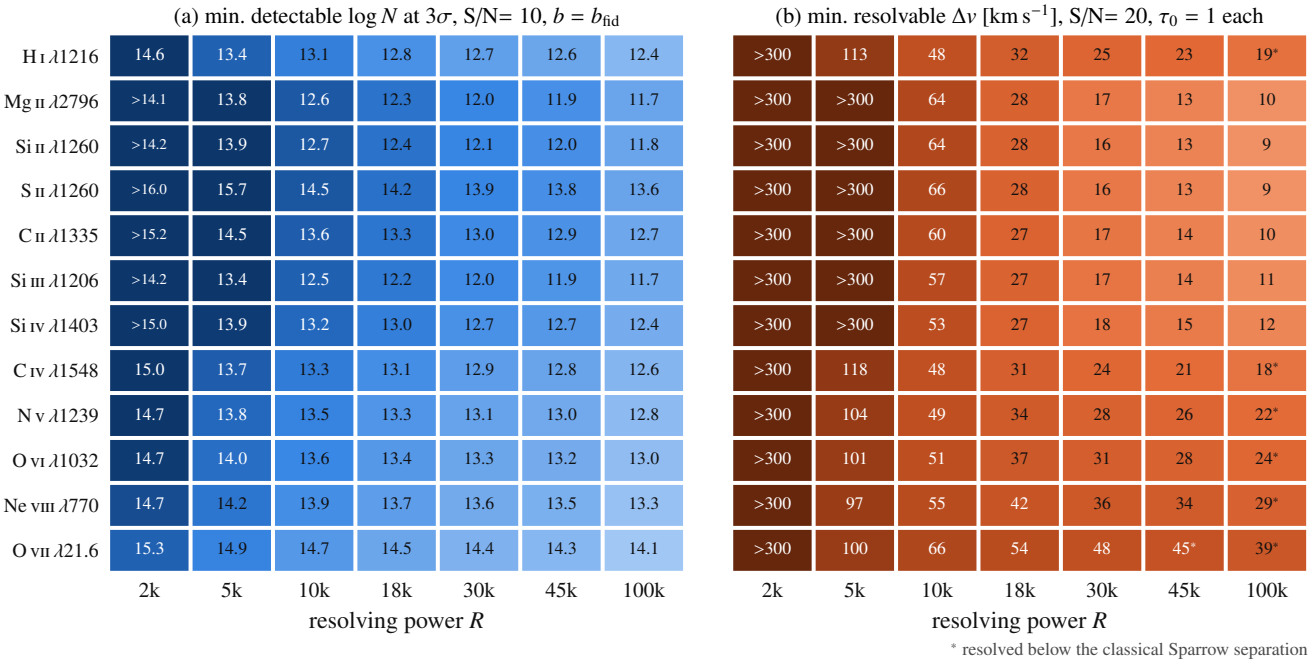

(a) min. detectable log N at 3σ, S/N= 10, $b = b_{\rm fid}$
H I λ1216 14.6 13.4 13.1 12.8 12.7 12.6 12.4
Mg II λ2796 >14.1 13.8 12.6 12.3 12.0 11.9 11.7
Si II λ1260 >14.2 13.9 12.7 12.4 12.1 12.0 11.8
S II λ1260 >16.0 15.7 14.5 14.2 13.9 13.8 13.6
C II λ1335 >15.2 14.5 13.6 13.3 13.0 12.9 12.7
Si III λ1206 >14.2 13.4 12.5 12.2 12.0 11.9 11.7
Si IV λ1403 >15.0 13.9 13.2 13.0 12.7 12.7 12.4
C IV λ1548 15.0 13.7 13.3 13.1 12.9 12.8 12.6
N V λ1239 14.7 13.8 13.5 13.3 13.1 13.0 12.8
O VI λ1032 14.7 14.0 13.6 13.4 13.3 13.2 13.0
Ne VIII λ770 14.7 14.2 13.9 13.7 13.6 13.5 13.3
O VII λ21.6 15.3 14.9 14.7 14.5 14.4 14.3 14.1
2k 5k 10k 18k 30k 45k 100k
resolving power R
(b) min. resolvable Δv [km s^{-1}], S/N= 20, $\tau_0 = 1$ each
>300 113 48 32 25 23 19*
>300 >300 64 28 17 13 10
>300 >300 64 28 16 13 9
>300 >300 66 28 16 13 9
>300 >300 60 27 17 14 10
>300 >300 57 27 17 14 11
>300 >300 53 27 18 15 12
>300 118 48 31 24 21 18*
>300 104 49 34 28 26 22*
>300 101 51 37 31 28 24*
>300 97 55 42 36 34 29*
>300 100 66 54 48 45* 39*
2k 5k 10k 18k 30k 45k 100k
resolving power R
* resolved below the classical Sparrow separation